Graphical Demonstration of Heat Conduction in Cylindrical Rods Using Python

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Abstract:

This project is a study on the behaviour of heat conduction in cylindrical conductors. A metallic conductor of standard values is taken for the purpose of demonstration of heat transfer in steady state conditions. Fourier's Law of Heat Conduction is applied and heat lost from the surface is considered with the help of Newton's Law of Cooling. It concludes with an equation matching the results of Forbes' Experiment, which shows that temperature decays with length along the conductor and approaches ambient temperature.

Introduction:

In Physics, temperature is a measure of the degree of hotness or coldness of a body with respect to a standard scale. It is a scalar quantity which increases when heat is supplied to a body and decreases when heat is taken away from a body (or cooled). Therefore, heat is the transfer of energy between two bodies or within a body that arises due to a temperature difference.

In a metallic conductor, when one end is heated (supplied with energy), the temperature of that region increases. Heat has an inherent property of flowing from the region of higher temperature to the region of lower temperature, that is, along the temperature gradient. Therefore, the temperature approaches room temperature or ambient temperature as we go along the length of the conductor.

In the 19th century, Joseph Fourier studied the phenomenon of heat transfer in conductors and concluded with a mathematical formula

relating heat flow with temperature gradient. Later, James David Forbes conducted experiments on conductors to study the decay of temperature across the length of the conductor. He heated the rod at one end and measured the temperature at different points of the conductor to conclude with a mathematical formula to determine the thermal conductivity of metals. This project follows a similar approach to demonstrate decay of temperature across the length of a conductor.

Cylindrical Metal Conductor:

Heat transfer from a region of higher temperature to a region of lower temperature occurs in three ways: Conduction, Convection and Radiation. In a metallic conductor, heat transfer happens through Conduction. It is the process of heat transfer that happens from one particle to another without disturbing their equilibrium position.

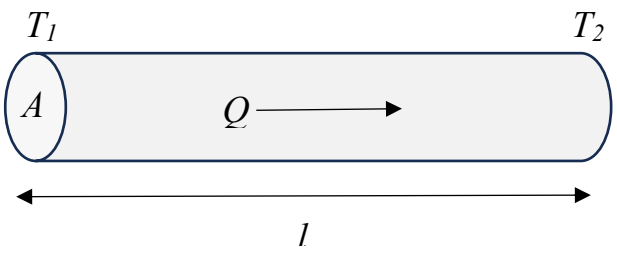
Consider a cylindrical metallic conductor of length l and area of cross section A. One end of the conductor is heated for time t. The temperature at the heated end is T_1 and the other end is T_2. The heat supplied to the conductor is Q and:

$Q \propto A$

$Q \propto t$

$Q \propto (T_1 - T_2)$

$Q \propto \frac{1}{l}$



Substituting everything into an equation with a proportionality constant K, we obtain:

$$Q = \frac{KA(T_1 - T_2)\mathrm{t}}{\mathrm{l}}$$

where, K is the Thermal Conductivity of the material of the conductor.

Consider a point at a length at distance x from the heated end of the conductor. Let T(x) be the temperature at the point x. Consider a small

element of the rod between x and x + dx. We can rewrite the expression as:

$$\frac{dQ}{dt} = -KA\left(\frac{dT}{dx}\right) \qquad \dots\dots\dots (1)$$

Where,

- $\frac{dQ}{dt}$ is the rate of flow of heat through the rod
- $\frac{dT}{dx}$ is the temperature gradient across the rod
- K is the coefficient of thermal conductivity of the material
- Negative sign represents flow of heat from higher temperature to lower temperature

Equation (1) represents Fourier's Law of Heat Conduction. This is the starting point of the project.

Temperature Variation Along The Length of the Rod:

When energy is supplied to the small element of the rod in the form of heat, some of the energy is lost to the surroundings as well. The energy balance can be represented as:

Heat Entering the Element = Heat Leaving the Element + Heat Lost to Surroundings

or: $Q_{in} = Q_{out} + Q_{lost}$ $\qquad \dots\dots\dots (2)$

1. Q_{in}:
   By Fourier's Law,
   $$Q_{in} = -KA\left(\frac{dT}{dx}\right)|_x \qquad \dots\dots\dots (3)$$

2. Q_{out}:
   By Fourier's Law,
   $$Q_{out} = -KA\left(\frac{dT}{dx}\right)|_{(x+dx)}$$

We can apply Taylor Series expansion for this function about x. Let us take $f(x) = \frac{dT}{dx}$ and expand it about x:

$$= f(x) + \frac{dx}{1!}.\frac{df'}{dx} + \frac{(dx)^2}{2!}.\frac{df''}{dx} + \cdots$$

Since the terms involving $\frac{df''}{dx}$ and higher degrees are very small, they can be neglected and hence:

$$= f(x) + dx\left(\frac{df'}{dx}\right)$$

Substituting for f(x):

$$= \frac{dT}{dx} + dx(\frac{d^2T}{dx^2})$$

$$Q_{out} = -KA\left(\frac{dT}{dx} + dx\left(\frac{d^2T}{dx^2}\right)\right) \ldots \ldots \ldots (4)$$

3. Q_{lost}:
   From Newton's Law of Cooling, we know that heat lost to the surroundings is directly proportional to the temperature difference between the conductor and surroundings, and to the surface area of the conductor.
   In the small element of the rod, we can denote the heat lost as:
   $$Q_{lost} = h(T - T_a)(2\pi r dx) \ldots \ldots \ldots (5)$$
   Where,
   - h is the heat transfer coefficient
   - T is the temperature of the conductor at point x
   - T_a is the ambient temperature
   - 2πrdx is the surface area of the element of length dx

Substituting equations (3), (4) and (5) in equation (2), we get:

$$-KA\left(\frac{dT}{dx}\right) = -KA\left(\frac{dT}{dx} + dx\left(\frac{d^2T}{dx^2}\right)\right) + h(T - T_a)(2\pi r dx)$$

$$= -KA\left(\frac{dT}{dx}\right) = -KA\left(\frac{dT}{dx}\right) - KA\left(\frac{d^2T}{dx^2}\right)dx + h(T - T_a)(2\pi r dx)$$

Cancelling out common terms:

$$= KA\left(\frac{d^2T}{dx^2}\right)dx = h(T - T_a)(2\pi r)dx$$

Cancel out dx and let 2πr = P (Perimeter of the element):

$$= KA\left(\frac{d^2T}{dx^2}\right) = hP(T - T_a)$$

$$= \frac{d^2T}{dx^2} = \frac{hP}{KA}(T - T_a) \ldots \ldots \ldots (6)$$

We now have a differential equation to solve.

Let $\frac{hP}{KA}$ = m and T - T_a = θ

$$= \frac{d\theta}{dx} = \frac{dT}{dx}$$

$$= \frac{d^2\theta}{dx^2} = \frac{d^2T}{dx^2}$$

Substituting in equation (6), we get:

$$= \frac{d^2\theta}{dx^2} = m\theta$$

$$= \frac{d^2\theta}{dx^2} - m\theta = 0$$

The Auxiliary Equation is:

$$= r^2 - m = 0$$

$$r = \pm\sqrt{m}$$

Therefore, the solution of the differential equation is:

$$\theta = C_1 e^{x\sqrt{m}} + C_2 e^{-x\sqrt{m}}$$

Substituting the value of θ:

$$T - T_a = C_1 e^{x\sqrt{m}} + C_2 e^{-x\sqrt{m}}$$

$$= T = T_a + C_1 e^{x\sqrt{m}} + C_2 e^{-x\sqrt{m}} \ldots \ldots \ldots (7)$$

Now we have to find the values of C_1 and C_2. We know that when the value of x approaches infinity, the T(x) = T becomes equal to T_a.

When x → ∞, T → T_a

Substituting in Equation (7), the value of $e^{x\sqrt{m}}$ becomes infinity, which is not possible. Therefore it can be concluded that $C_1 = 0$.

Equation (7) becomes: $T = T_a + C_2 e^{-x\sqrt{m}}$

When x = 0, T(x) = T_0 which is the temperature of the heated end of the rod. Substituting in equation (7), we get the value of C_2 as:

$$C_2 = T_0 - T_a$$

Finally substituting everything into the equation, we obtain:

$$\boldsymbol{T - T_a = (T_0 - T_a)e^{-x\sqrt{m}}} \quad \ldots\ldots\ldots (8)$$

Where, $m = \frac{hP}{KA}$

This is the final derived equation which matches the results of James David Forbes' experiment on a heated cylindrical conductor. The decay in temperature can be demonstrated graphically using Python.

<u>Graphical Demonstration:</u>

The temperature distribution across the length of a solid and hollow conductor can be demonstrated graphically on Python.

1. Solid Rod:
   Consider a solid copper conductor of length 1 metre with the following values:
   - Thermal Conductivity K = 401 W m^{-1} K^{-1}
   - Heat Transfer Coefficient h = 10 W m^{-2} K^{-1}
   - Radius r = 0.01 m

   Area of Cross Section A = πr^2

   Perimeter P = $2\pi r$

2. Hollow Rod:
   Consider a hollow copper conductor of length 1 metre with the following values:
   - Thermal Conductivity K = 401 W m^{-1} K^{-1}
   - Heat Transfer Coefficient h = 10 W m^{-2} K^{-1}
   - Outer Radius R = 0.01 m
   - Inner Radius r = 0.005 m

   Area of Cross Section $A = \pi(R^2 - r^2)$

   Perimeter $P = 2\pi R$

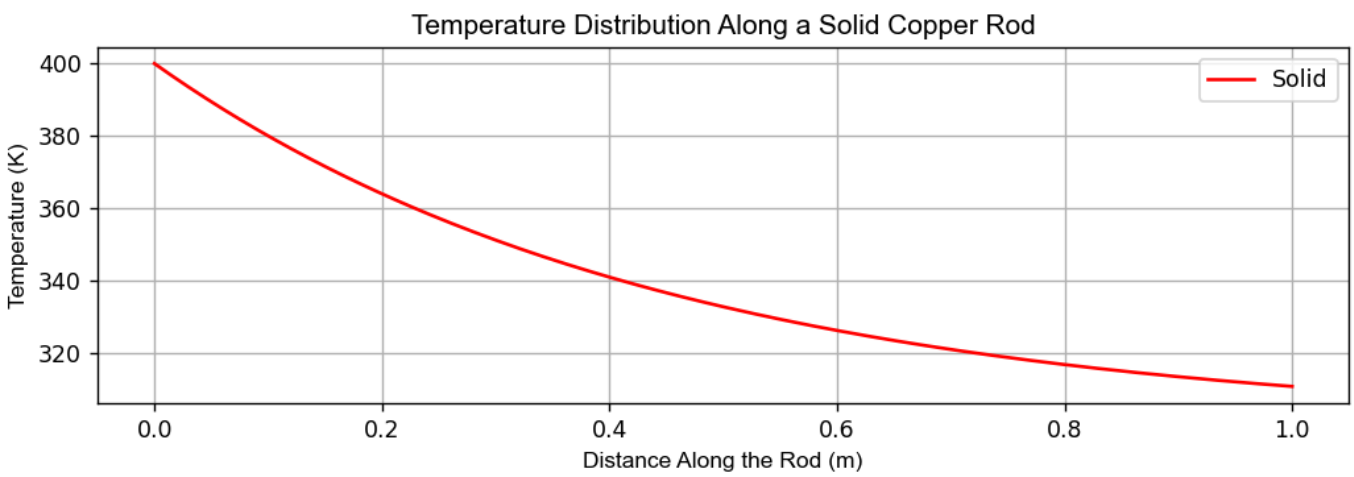
Temperature of Heated End T_0 = 400 K

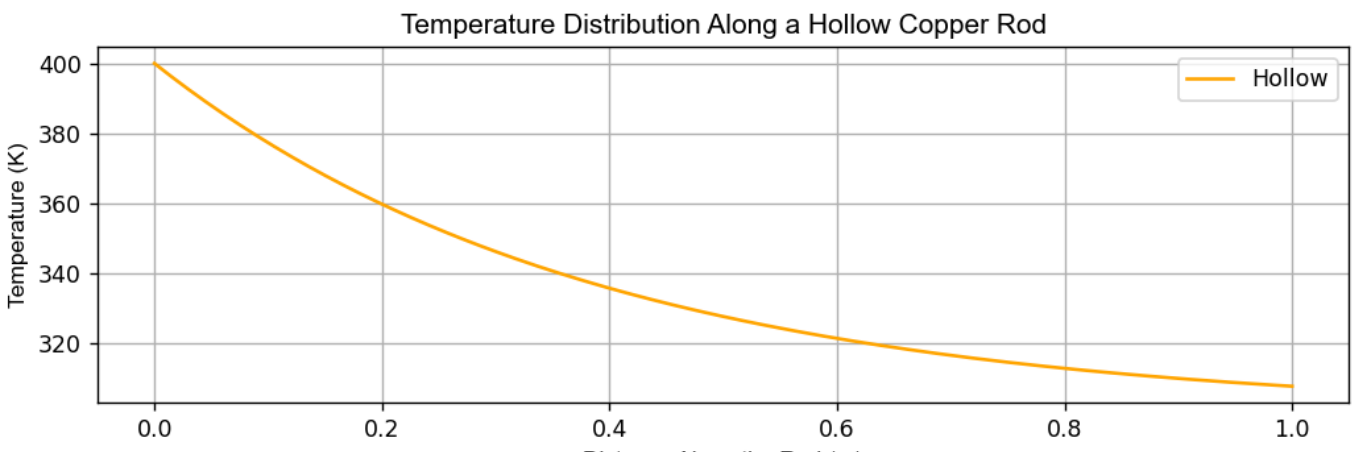
Ambient Temperature T_a = 300 K

Substitute the values in Equation (8) and plot a graph from x=0 to x=1 (m) that is, across the length of the conductors:

$$T - T_a = (T_0 - T_a)e^{-x\sqrt{m}}$$

The following graphs are obtained:





It can be observed that the temperature decays as the length of the conductor is varied from 0 to 1, and approaches ambient temperature T_a from the temperature at the heated end $T_0 = 400$ K.

It can also be observed that the hollow rod shows a faster decrease in temperature as compared to the solid rod. This is because since the area exposed to air is more, the heat lost to surroundings is more as well.

Conclusion:

It was concluded that when one end of a cylindrical conductor is heated, the temperature decays exponentially along the length of the conductor and approaches ambient temperature. The mathematical expression that matches Forbes' Experiment on conduction of heat along a metallic conductor was derived with the help of Fourier's Law of Heat Conduction and Newton's Law of Cooling. The result was graphically demonstrated for two cases of i) A Solid Copper Conductor and ii) A Hollow Copper Conductor, using Python. The result showed coherence with the conclusion of the experiment conducted by Forbes.

Appendix A - Python Code:

```
import numpy as np
import matplotlib.pyplot as plt
k = 401 # W /(m K)
h = 10 # W / (m^2 K)
T0 = 400 # K
Ta = 300 # K

r_solid = 0.01 # m
R_hollow = 0.01 # m
r_hollow = 0.005 # m

A_solid = np.pi * r_solid**2 # m^2
A_hollow = np.pi * (R_hollow**2 - r_hollow**2) # m^2

P_solid = 2 * np.pi * r_solid # m
P_hollow = 2 * np.pi * R_hollow # m

m_solid = (h * P_solid)/(k * A_solid) # m^-2
m_hollow = (h * P_hollow)/(k * A_hollow) # m^-2

x = np.linspace(0, 1, 200) # m

T_solid = Ta + ((T0 - Ta) * np.exp(-np.sqrt(m_solid)*x)) # K
T_hollow = Ta + ((T0 - Ta) * np.exp(-np.sqrt(m_hollow)*x)) # K

figure, axes = plt.subplots(2, 1)
axes[0].plot(x, T_solid, color="red")
axes[0].set_title("Temperature Distribution Along a Solid Copper Rod",
family = "arial")
axes[0].set_xlabel("Distance Along the Rod (m)", family = "arial")
axes[0].set_ylabel("Temperature (K)", family = "arial")
axes[0].legend(["Solid"])
axes[0].grid(True)

axes[1].plot(x, T_hollow, color="orange")
axes[1].set_title("Temperature Distribution Along a Hollow Copper Rod",
family = "arial")
axes[1].set_xlabel("Distance Along the Rod (m)", family = "arial")
axes[1].set_ylabel("Temperature (K)", family = "arial")
axes[1].legend(["Hollow"])
axes[1].grid(True)

plt.tight_layout()
plt.savefig("temperature_distribution.png", dpi=300)
plt.show()
```

References:

- Joseph Fourier, The Analytical Theory of Heat (19th Century)
- James David Forbes, Experimental studies on heat conduction in metals (19th Century)
- The Feynman Lectures on Physics, Volume 1
- NumPy and Matplotlib